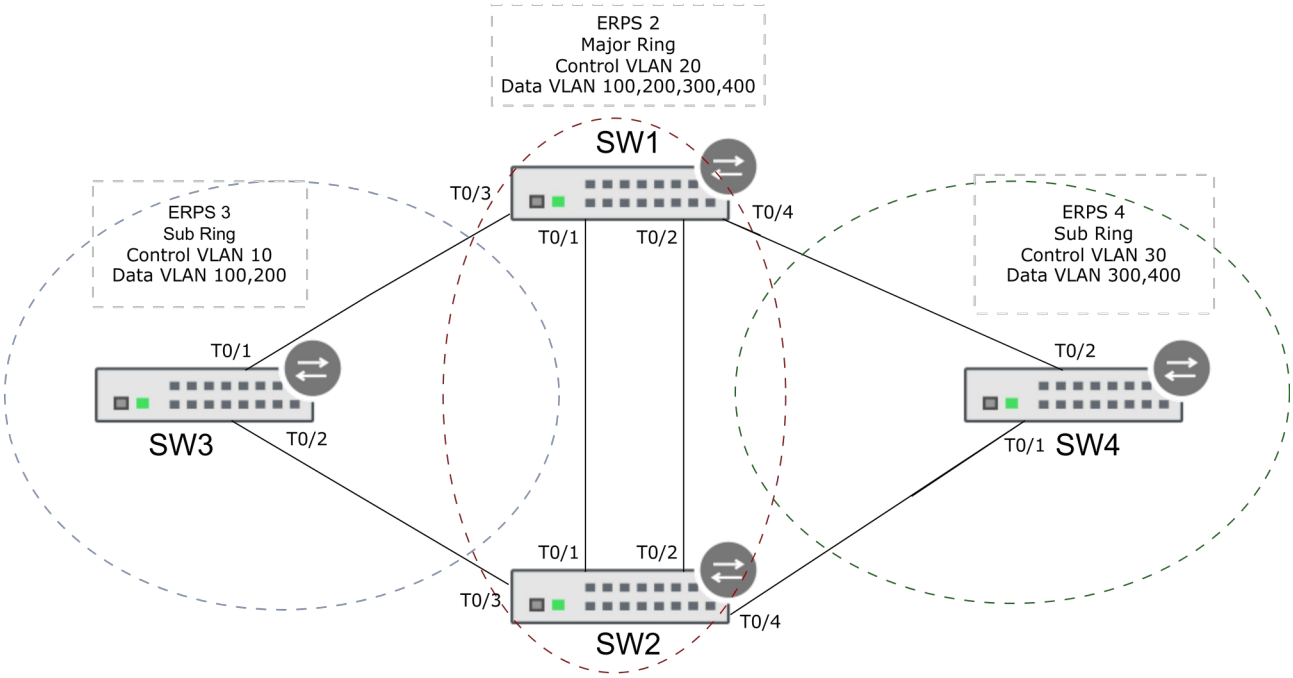
Test Report
on

# Multi-ring ERPS
# on
# BDCOM S5828

| | |
|---|---|
| Objective | To check ERPS in multi-ring topology with S5828. |
| Testing environment | Lab environment at BDCOM Asia Pacific Office, Dhaka, Bangladesh |
| Tested By | Salakin, After Sales Engineer |
| Date | 31st January 2023 |
| Testing devices | 1. 4 units of S5828<br>Firmware version : Version 2.2.0D Build 108375 |

# Topology:




# Configuration:

## SW1:

```
Switch_config#hostname SW1
SW1_config#no spanning-tree#
SW1_config#vlan 1,10,20,30,100,200,300,400

SW1_config#erps 2
SW1_config_ring2#interconnection-node
SW1_config_ring2#major-ring
SW1_config_ring2#control-vlan 20
SW1_config_ring2#exit
```

```
SW1_config#erps 3
SW1_config_ring3#interconnection-node
SW1_config_ring3#sub-ring
SW1_config_ring3#control-vlan 10
SW1_config_ring3#exit

SW1_config#erps 4
SW1_config_ring4#interconnection-node
SW1_config_ring4#sub-ring
SW1_config_ring4#major-ring-id 2
SW1_config_ring4#control-vlan 30
SW1_config_ring4#exit

SW1_config#interface TGigaEthernet0/1
SW1_config_tg0/1#switchport mode trunk
SW1_config_tg0/1#erps 2 rpl
SW1_config_tg0/1#exit

SW1_config#interface TGigaEthernet0/2
SW1_config_tg0/2#switchport mode trunk
SW1_config_tg0/2#erps 2 ring-port
SW1_config_tg0/2#exit

SW1_config#interface TGigaEthernet0/3
SW1_config_tg0/3#switchport mode trunk
SW1_config_tg0/3#erps 3 rpl
SW1_config_tg0/3#exit

SW1_config#interface TGigaEthernet0/4
SW1_config_tg0/4#switchport mode trunk
SW1_config_tg0/4#erps 4 ring-port#
SW1_config_tg0/4#exit
```

## SW2:

```
Switch_config#hostname SW2
SW2_config#no spanning-tree
SW2_config#vlan 1,10,20,30,100,200,300,400

SW2_config#erps 2
SW2_config_ring2#interconnection-node
SW2_config_ring2#major-ring
SW2_config_ring2#control-vlan 20
SW2_config_ring2#exit

SW2_config#erps 3
SW2_config_ring3#interconnection-node
SW2_config_ring3#sub-ring
```

```
SW2_config_ring3#control-vlan 10
SW2_config_ring3#exit

SW2_config#erps 4
SW2_config_ring4#interconnection-node
SW2_config_ring4#sub-ring
SW2_config_ring4#major-ring-id 2
SW2_config_ring4#control-vlan 30
SW2_config_ring4#exit

SW2_config#interface TGigaEthernet0/1
SW2_config_tg0/1#switchport mode trunk
SW2_config_tg0/1#erps 2 ring-port
SW2_config_tg0/1#exit

SW2_config#interface TGigaEthernet0/2
SW2_config_tg0/2#switchport mode trunk
SW2_config_tg0/2#erps 2 ring-port
SW2_config_tg0/2#exit

SW2_config#interface TGigaEthernet0/3
SW2_config_tg0/3#switchport mode trunk
SW2_config_tg0/3#erps 3 ring-port
SW2_config_tg0/3#exit

SW2_config#interface TGigaEthernet0/4
SW2_config_tg0/4#switchport mode trunk
SW2_config_tg0/4#erps 4 rpl
SW2_config_tg0/4#exit
```

## SW3:

```
Switch_config#hostname SW3
SW3_config#no spanning-tree
SW3_config#vlan 1,10,100,200

SW3_config#erps 3
SW3_config_ring3#sub-ring
SW3_config_ring3#control-vlan 10
SW3_config_ring3#exit

SW3_config_#interface TGigaEthernet0/1
SW3_config_tg0/1#switchport mode trunk
SW3_config_tg0/1#erps 3 ring-port
SW3_config_tg0/1#exit

SW3_config#interface TGigaEthernet0/2
SW3_config_tg0/2#switchport mode trunk
```

```
SW3_config_tg0/2#erps 3 ring-port
SW3_config_tg0/2#exit
```

## SW4:

```
Switch_config#hostname SW4
SW4_config#no spanning-tree
SW4_config#vlan 1,30,300,400

SW4_config#erps 4
SW4_config_ring4#sub-ring
SW4_config_ring4#control-vlan 30
SW4_config_ring4#exit

SW4_config#interface TGigaEthernet0/1
SW4_config_tg0/1#switchport mode trunk
SW4_config_tg0/1#erps 4 ring-port
SW4_config_tg0/1#exit

SW4_config#interface TGigaEthernet0/2
SW4_config_tg0/2#switchport mode trunk
SW4_config_tg0/2#erps 4 ring-port
SW4_config_tg0/2#exit
```

# Results:

## SW1:

```
SW1#show erps

Ethernet Ring Protection Switching

Ring2
             This node is the RPL Owner
  Node ID    Role           RPL
             Address        9845.6243.4812
             Control Vlan   20
             Version        2
             This node is the interconnection node
             This ring is a major-ring
             Ring ID validy check:    True
             RAPS Virtual Channel:    True
             Revertive Mode:    Revertive
             State  Idle            WTR  True
             Signal Fail  False     Sending  Unknown
             WTR time 0/20 sec  WTB time 0/6 sec
             Guard time 0/500 ms  Send time 5/5 sec
```

```
Status code: D - LINK DOWN, F - SIGNAL FAIL
Interface        Role           State        Status       MEP Role
---------------- -------------- ------------ ------------ ---------------
tg0/1            RPL            BLK                       NULL
tg0/2            Ring-Port      FWD                       NULL

Ring3
             This node is the RPL Owner
  Node ID    Role           RPL
             Address        9845.6243.4812
             Control Vlan   10
             Version        2
             This node is the interconnection node
             This ring is a sub-ring
             This sub-ring shall be connected to a major-ring.
             Ring ID validy check:    True
             RAPS Virtual Channel:    True
             Revertive Mode:    Revertive
             State  Idle            WTR  True
             Signal Fail  False     Sending  NR-RB(DNF)
             WTR time 0/20 sec  WTB time 0/6 sec
             Guard time 0/500 ms  Send time 4/5 sec

Status code: D - LINK DOWN, F - SIGNAL FAIL
Interface        Role           State        Status       MEP Role
---------------- -------------- ------------ ------------ ---------------
tg0/3            RPL            BLK          NULL         NULL

Ring4
  Node ID    Role           Normal
             Address        9845.6243.4812
             Control Vlan   30
             Version        2
             This node is the interconnection node
             This ring is a sub-ring
             This sub-ring is connected to the major-ring: 2
             Ring ID validy check:    True
             RAPS Virtual Channel:    True
             Revertive Mode:    Revertive
             State  Idle
             Signal Fail  False     Sending  None
             WTR time 0/20 sec  WTB time 0/6 sec
             Guard time 0/500 ms  Send time 0/5 sec

Status code: D - LINK DOWN, F - SIGNAL FAIL
Interface        Role           State        Status       MEP Role
---------------- -------------- ------------ ------------ ---------------
tg0/4            Ring-Port      FWD          NULL         NULL
```

## SW2:

```
SW2#show erps

Ethernet Ring Protection Switching

Ring2
  Node ID    Role           Normal
             Address        9845.6243.E08A
             Control Vlan   20
             Version        2
             This node is the interconnection node
             This ring is a major-ring
             Ring ID validy check:    True
             RAPS Virtual Channel:    True
             Revertive Mode:    Revertive
             State  Idle
             Signal Fail  False     Sending  None
             WTR time 0/20 sec  WTB time 0/6 sec
             Guard time 0/500 ms  Send time 0/5 sec

Status code: D - LINK DOWN, F - SIGNAL FAIL
Interface        Role           State        Status       MEP Role
---------------- -------------- ------------ ------------ ---------------
tg0/1            Ring-Port      FWD                       NULL
tg0/2            Ring-Port      FWD                       NULL

Ring3
  Node ID    Role           Normal
             Address        9845.6243.E08A
             Control Vlan   10
             Version        2
             This node is the interconnection node
             This ring is a sub-ring
             This sub-ring shall be connected to a major-ring.
             Ring ID validy check:    True
             RAPS Virtual Channel:    True
             Revertive Mode:    Revertive
             State  Idle
             Signal Fail  False     Sending  None
             WTR time 0/20 sec  WTB time 0/6 sec
             Guard time 0/500 ms  Send time 0/5 sec

Status code: D - LINK DOWN, F - SIGNAL FAIL
Interface        Role           State        Status       MEP Role
---------------- -------------- ------------ ------------ ---------------
```

```
tg0/3            Ring-Port      FWD          NULL         NULL


Ring4
             This node is the RPL Owner
  Node ID    Role           RPL
             Address        9845.6243.E08A
             Control Vlan   30
             Version        2
             This node is the interconnection node
             This ring is a sub-ring
             This sub-ring is connected to the major-ring: 2
             Ring ID validy check:    True
             RAPS Virtual Channel:    True
             Revertive Mode:    Revertive
             State  Idle            WTR  True
             Signal Fail  False     Sending  NR-RB(DNF)
             WTR time 0/20 sec  WTB time 0/6 sec
             Guard time 0/500 ms  Send time 2/5 sec

Status code: D - LINK DOWN, F - SIGNAL FAIL
Interface        Role           State        Status       MEP Role
---------------- -------------- ------------ ------------ ---------------
tg0/4            RPL            BLK           NULL         NULL
```

## SW3:

```
SW3#show erps

Ethernet Ring Protection Switching

Ring3
  Node ID    Role           Normal
             Address        9845.6243.E180
             Control Vlan   10
             Version        2
             This ring is a sub-ring
             Ring ID validy check:    True
             RAPS Virtual Channel:    True
             Revertive Mode:    Revertive
             State  Idle
             Signal Fail  False     Sending  None
             WTR time 0/20 sec  WTB time 0/6 sec
             Guard time 0/500 ms  Send time 0/5 sec

Status code: D - LINK DOWN, F - SIGNAL FAIL
Interface        Role           State        Status       MEP Role
```

```
---------------- -------------- ------------ ------------ ---------------
tg0/1            Ring-Port      FWD                       NULL
tg0/2            Ring-Port      FWD                       NULL
```

## SW4:

```
SW4#show erps

Ethernet Ring Protection Switching

Ring4
  Node ID    Role           Normal
             Address        9845.6243.4626
             Control Vlan   30
             Version        2
             This ring is a sub-ring
             Ring ID validy check:    True
             RAPS Virtual Channel:    True
             Revertive Mode:    Revertive
             State  Idle
             Signal Fail  False     Sending  None
             WTR time 0/20 sec  WTB time 0/6 sec
             Guard time 0/500 ms  Send time 0/5 sec

Status code: D - LINK DOWN, F - SIGNAL FAIL
Interface        Role           State        Status       MEP Role
---------------- -------------- ------------ ------------ ---------------
tg0/1            Ring-Port      FWD                       NULL
tg0/2            Ring-Port      FWD                       NULL
```